Mock Paper A(2)

Started on	Thursday, 21 June 2018, 8:31 AM
State	Finished
Completed on	Thursday, 21 June 2018, 8:32 AM
Time taken	18 secs
Grade	**0.00** out of 100.00
Feedback	You are on the right track. Could you spend some more time on your revision?

Question 1
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 052
Nomifensine is an antidepressant that has noradrenaline, dopamine and also serotonin reuptake inhibition properties. It has been withdrawn due to its propensity to cause which of the following side effects?

Select one:
- Haemolytic anaemia
- Agranulocytosis
- Seizures
- Thrombocytopenia
- Hepatitis

Your answer is incorrect.

Nomifensine has noradrenaline, dopamine and serotonin reuptake inhibition properties. It was withdrawn due to haemolytic anaemia in a small number of patients. It was a particularly effective antidepressant for patients with epilepsy as, it did not alter the seizure threshold.

The correct answer is: Haemolytic anaemia

Question 2
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 053
Which of the following medications is preferred option to treat a 75 year old man with a relapse of psychotic depression and has been taking treatment for prostatic hypertrophy in addition to high cholesterol?

Select one:
- Thioridazine
- Chlorpromazine
- Haloperidol
- Olanzapine
- Clozapine

Your answer is incorrect.

High-potency neuroleptics, such as haloperidol have lower anticholinergic effects and less likely to cause urinary retention compared to low-potency medications such as chlorpromazine in elderly patients with prostatic hypertrophy. Clozapine is not recommended because of its powerful anticholinergic effects, its tendency to cause hypotension, and the risk for agranulocytosis. Thioridazine can cause fatal arrhythmias by prolonging the QT interval, in addition to the high anticholinergic profile. Finally, olanzapine may not be appropriate due to cholesterol related issues.

The correct answer is: Haloperidol

Question 3
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 054
A bipolar disorder patient on lithium treatment developed signs of severe lithium toxicity. What is the best immediate treatment option?

Select one:
- Peritoneal dialysis
- Forced alkaline diuresis
- Hemodialyisis
- Volume repletion
- Arteriovenous hemodynamic filtration

Your answer is incorrect.

Moderate and severe lithium toxicity require aggressive approaches. Initial general antipoisoning measures like gastric lavage may be helpful, but ultimate success depends upon elimination of lithium from the body. Hemodialysis will rapidly eliminate lithium from the body.

The correct answer is: Hemodialyisis

Question 4
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 055
Tardive dyskinesia induced by psychotropics is attributed to which of the following mechanisms?

Select one:
- Low D-2 receptor occupancy
- High D-2 receptor occupancy

- High D-2 receptor sensitivity
- Low D-1 receptor sensitivity
- Low D-2 receptor sensitivity

Your answer is incorrect.

Theories explaining tardive dyskinesia include dopaminergic/cholinergic imbalance, upregulation/supersensitivity of postsynaptic D2 receptors in the basal ganglia following chronic blockade and GABA hypofunction leading to enhanced DA transmission.

The correct answer is: High D-2 receptor sensitivity

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 056

A 67-year-old gentleman has been on treatment for depression. He went to the local beach for sunbathing on a summer day afternoon and presented to A and E complaining of nausea, dizziness and leg cramps.

Select one:

- Hypoglycemia
- Hyponatraemia
- Hyperkalemia
- Hypokalemia
- Hypernatraemia

Your answer is incorrect.

Most antidepressants are strongly linked with hyponatraemia. The clinical features of hyponatraemia include nausea, dizziness, lethargy, confusion, cramps and seizures. Other risk factors, in this case, includes old age, warm weather and drug treatments such as diuretics.

The correct answer is: Hyponatraemia

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 057

A 65 year old woman who is being treated for depression comes to the Medical Emergency Unit with lethargy, cramps, poor memory and weight loss on a hot afternoon. She is taking some unknown psychotropic medication for a long time. Also, more recently, she started taking 'water tablets' for swelling in her legs. Which of the following is a likely cause of her symptoms?

Select one:

- Hypokalemia
- Hypernatremia
- Hyponatremia
- Neuroleptic malignant syndrome
- Serotonin syndrome

Your answer is incorrect.

The story is suggestive of hyponatremia induced by diuretics and antidepressants.

The correct answer is: Hyponatremia

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 060

What is the main risk involved in treating a patient with combination of clozapine and carbamazepine?

Select one:

- Neuroleptic malignant syndrome
- Diabetes mellitus
- Seizures
- Leucopenia
- Weight gain

Your answer is incorrect.

Clozapine can cause neutropenia. Carbamazepine causes a chronic low white blood cell count.

The correct answer is: Leucopenia

Question 8

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 061

Whilst having ECT, which of the following medications may require lower than normal current to be selected?

Select one:

- Sodium valproate
- Nitrazepam
- Lithium
- Zolpidem
- Lamotrigine

Your answer is incorrect.

Lithium could reduce seizure threshold even on normal therapeutic doses. So patients on Lithium, while having ECT may require lower than normal current.

The correct answer is: Lithium

Question 9
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 062
Which of the following agents is likely to have caused a newborn to have a defect of the tricuspid valve?

Select one:
- Valproate
- Paroxetine
- Lithium
- Zopiclone
- Naltrexone

Your answer is incorrect.

Valproate induced neural tube effects are predominantly lumbosacral, but anencephaly is also seen occasionally. Tricuspid defects are often seen in lithium-treated patients, with extreme cases having full-blown Ebstein's anomaly.

The correct answer is: Lithium

Question 10
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 064
Which one among the following drug usage has been associated with hyperparathyroidism?

Select one:
- Sodium Valproate
- Gabapentin
- Carbamazepine
- Lithium
- Lamotrigine

Your answer is incorrect.

Hypocalcaemia and hyperparathyroidism are uncommon side effects associated with lithium therapy. The common medical symptoms include nausea, polydipsia and constipation.

The correct answer is: Lithium

Question 11
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 067
A depressed patient experiences constipation and dry mouth whilst on amitriptyline. This is mainly due to

Select one:
- M1 receptor blockade
- Alpha-1 receptor blockade
- Alpha 1 agonism
- M 2 receptor blockade
- M3 receptor blockade

Your answer is incorrect.

Amitriptyline and other tricyclic antidepressants can induce anticholinergic side effects such as constipation, dry mouth and urinary retention. Muscarinic receptor blockade leads to anticholinergic side effects, especially through M3 receptor blockade.

The correct answer is: M3 receptor blockade

Question 12
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 069
Which of the following antidepressants is associated with a significant risk of agranulocytosis compared to the others listed?

Select one:
- Mirtazapine
- Bupropion
- Escitalopram
- Fluoxetine
- Venlafaxine

Your answer is incorrect.

All drugs ending with a suffix -zapine can cause agranulocytosis (mirtazapine, olanzapine, clozapine, carbamazepine, etc.). In addition, mianserin can also cause this effect

The correct answer is: Mirtazapine

Question 13
Not answered

MoK Adverse Effects 070
Which of the following antidepressants has a recognised side effect of elevated cholesterol?

Marked out of 1.00

Flag question

Select one:

- fluoxetine
- imipramine
- venlafaxine
- paroxetine
- mirtazapine

Your answer is incorrect.

Mirtazapine - similar to olanzapine - is associated with blood dyscrasias and hyperlipidemia.

The correct answer is: mirtazapine

Question 14

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 071
Which of the following is NOT a dystonia?

Select one:

- Writer's cramp
- Oculogyric crisis
- Motor tic
- Torticollis
- Blepharospasm

Your answer is incorrect.

Tics are not classified as dystonia

The correct answer is: Motor tic

Question 15

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 072
Which one among the following is not a feature of SSRI discontinuation syndrome

Select one:

- Increased dreams
- Dizziness
- Insomnia
- Myoclonic Jerks
- Electric shock like sensations

Your answer is incorrect.

SSRI discontinuation syndrome: Sensory abnormalities include electric shock-like sensations, symptoms of a generalised disequilibrium including dizziness, insomnia, increased dreams, and general somatic symptoms such as headache, lethargy, gastrointestinal symptoms and mood changes such as anxiety, agitation and depression.

The correct answer is: Myoclonic Jerks

Question 16

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 030
Which of the following agents exert an allosteric action on the nicotinic receptors?

Select one:

- Memantine
- Gallantamine
- Rivastigmine
- Tacrine
- Donepezil

Your answer is incorrect.

The two most common approaches to correcting cholinergic deficits in dementia are to increase the synaptic availability of acetylcholine (ACh) by inhibiting acetylcholinesterase (AChE), or to mimic the effects of ACh by acting directly on nicotinic receptors. AChE inhibitors produce only short-term symptomatic improvement while long-term use of nicotinic agonists desensitizes nicotinic receptors limiting the duration of efficacy. Allosteric modulation of nAChR is a novel approach, which circumvents the development of tolerance. Allosteric modulators bind to a site on nAChR that is different to the binding site of the natural agonist, ACh. This allosteric interaction amplifies the actions of ACh at post- and presynaptic nAChR. In particular, presynaptic nAChR are capable of modulating the release of ACh and other neurotransmitters, such as glutamate, serotonin and GABA, which may contribute to symptoms of the illness. Allosteric modulation of nAChR could, therefore, produce significant therapeutic benefit in AD. As well as modulating nAChR, galantamine inhibits AChE. The extent to which the clinical benefits of galantamine are attributable specifically to its nicotinic effects is uncertain and requires further investigation (Excerpt retrieved from Maelick A., Dement Geriatr Cogn Disord 2000 Sep;11 Suppl 1:11-80).

The correct answer is: Gallantamine

Question 17

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 031
Antipsychotic drug induced akathisia is due to;

Select one:

- Dopamine D2 receptor activation in the hippocambus
- Dopamine receptor supersensitivity

- Low dopamine D2 receptor occupancy in striatum
- High dopamine D2 receptor occupancy in striatum
- Serotonin 5-HT2 receptor blockade

Your answer is incorrect.

Antipsychotic drug-induced akathisia is thought to be due to high dopamine D2 receptor occupancy in the striatum. Higher doses of antipsychotic medications are correlated with higher D2 receptor occupancy and worse subjective experience of akathisia. Most extrapyramidal symptoms are related to D2 receptor occupancy and, in particular, akathisia has an important subjective component.

The correct answer is: High dopamine D2 receptor occupancy in striatum

Question 18
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 032
A patient with depression is being treated by antidepressant medications. Which of the following changes may be observed?

Select one:
- Increasing corticotropin-releasing hormone mRNA
- Increasing cortisol
- Enhanced activity of HPA axis
- Increasing adrenocorticotropic hormone
- Increasing expression of corticosteroid receptors in brain

Your answer is incorrect.

Antidepressants reduce HPA axis activity by increasing negative feedback from steroid hormone.. This is made possible via increased corticosteroid receptors.

The correct answer is: Increasing expression of corticosteroid receptors in brain

Question 19
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 033
Which of the following is the most likely mode of action of disulfiram?

Select one:
- Inhibition of sulphuric acid
- Inhibition of alcohol decarboxylase
- Inhibition of alcohol dehyrogenase
- Inhibition of acetic acid
- Inhibition of aldehyde dehyrogenase

Your answer is incorrect.

Disulfiram causes an accumulation of acetaldehyde by blocking oxidation of alcohol after drinking. This action is mediated via the inhibition of hepatic aldehyde dehydrogenase enzyme. Upon alcohol consumption while taking antabuse, the patient experiences colicky abdominal pain, flushing, anxiety, dizziness, tachycardia, vomiting and headache. Symptoms are generally acute in onset, starting 5-15 minutes after drinking alcohol and last for several hours.

The correct answer is: Inhibition of aldehyde dehyrogenase

Question 20
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 034
Which symptom is acknowledged as part of cocaine withdrawal?

Select one:
- Intense craving
- Increased energy
- Euphoria
- Diminished need for sleep
- Formication

Your answer is incorrect.

Cocaine Withdrawal: Effects of cocaine are short-lived - rapid cessation occurs due to rapid metabolism - this can lead to repeated use. Dysphoria (crash), anhedonia, anxiety, irritability hypersomnolence, and sometimes agitation usually end within 18 hours though in heavy dependent users this can last up to a week, peaking in 3 days. The characteristic feature is intense cravings for cocaine with a striking lack of much physical withdrawal symptoms.

The correct answer is: Intense craving

Question 21
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 035
With respect to the neurotransmitter glutamate, which of the following is true?

Select one:
- It is an inhibitory neurotransmitter
- All receptors are ionotropic
- It acts via chloride channels
- It acts via NMDA receptors
- It is synthesized by the enzyme glutamate decarboxylase

Your answer is incorrect.

Glutamate decarboxylase (GAD) is an enzyme that catalyzes the decarboxylation of glutamate to GABA - thus GABA, not Glutamate, is synthesized by GAD.

The correct answer is: It acts via NMDA receptors

Question 22

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 036

Xanomeline has been advocated as a novel approach to treat schizophrenia. It acts via

Select one:

- Muscarinic antagonism
- D4 partial agonism
- 5HT7 blocakde
- D4 agonism
- M1/M4 agonism

Your answer is incorrect.

Xanomeline is undergoing Phase 3 trials as a treatment option for schizophrenia. It acts via M1/M4 agonism. http://ajp.psychiatryonline.org/cgi/content/full/165/8/1033

The correct answer is: M1/M4 agonism

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 037

Tom is on clozapine for treatment resistant psychosis. He is embarrassed about hypersalivation and wet-pillows every morning. The most likely mechanism of this adverse event is

Select one:

- M4 receptor agonism and alpha 2 receptor antagonism
- M4 receptor agonism and alpha 1 receptor antagonism
- M4 receptor antagonism and alpha 2 receptor antagonism
- M4 receptor antagonism and alpha 2 receptor antagonism
- M4 receptor antagonism and alpha 1 receptor antagonism

Your answer is incorrect.

Clozapine increases salivation through its muscarinic M4 receptor activation and/or blockade of alpha2-adrenoceptors, or by causing a distortion in swallowing reflex (possibly via M3). The M1 and M4 receptor subtypes predominate in various secretory glands. Clozapine is antagonistic at M1, M2, M3, and M5 receptor subtypes, but in vitro findings indicate that it is a full agonist at the M4 receptor. As a result, clozapine's stimulation of the M4 receptors in salivary glands leads to an increase in secretions. Agonism at the M4 receptor seems unique to clozapine and does not occur with other atypical antipsychotics. Treatment options include chewing gum, reducing the dosage of clozapine, or prescribing pharmacologic agents such as anticholinergics or alpha2-adrenoceptor agonists. (Ref; Rogers et al.,Clozapine-induced hypersalivation, Pharmacotherapy. 2000; 20(9): http://www.ncbi.nlm.nih.gov/pubmed/10852096)

The correct answer is: M4 receptor agonism and alpha 2 receptor antagonism

Question 24

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 039

Which antidepressant is a presynaptic alpha-2 antagonist and now classified as NaSSA?

Select one:

- Bupropion
- Venlafaxine
- Mirtazapine
- Reboxetine
- Buspirone

Your answer is incorrect.

Mirtazapine is a presynaptic alpha-2 antagonist that increases central neurotransmission by noradrenaline and serotonin. It is classified as Noradrenergic and specific serotonergic antagonist- NaSSA.

The correct answer is: Mirtazapine

Question 25

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 040

Which one among the following has very limited potential for causing a hypertensive crisis?

Select one:

- Cocaine
- Phenelzine
- Isocarboxazid
- Tranylcypromine
- Moclobemide

Your answer is incorrect.

Tyramine interactions are rare and mild with moclobemide but possible if high doses (more than 600 mg/day) used or if large quantities of tyramine ingested

The correct answer is: Moclobemide

Question 26
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 033
Which of the following is acceptable to be taken along with MAO inhibitors?

Select one:
- Mature cheese
- Broad bean pods
- Red wine
- Pickled Herring
- Grapefruit juice

Your answer is incorrect.
Except grapefruit juice, the others on the list are rich in tyramine and so can produce cheese reaction.
The correct answer is: Grapefruit juice

Question 27
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 039
Which routes of administration leads to nicotine reaching the brain fast?

Select one:
- Sublingual
- Inhalational
- Intravenous
- Cutaneous
- Oral

Your answer is incorrect.
Drugs administered via the inhalational route reaches the brain quickly.
The correct answer is: Inhalational

Question 28
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 040
Which one of the following statements about reboxetine is correct?

Select one:
- It is highly cardiotoxic.
- Reboxetine is a hepatic enzyme inhibitor
- Reboxetine is a hepatic enzyme inducer
- It does not affect hepatic cytochrome P450 enzymes.
- It is completely excreted via kidneys

Your answer is incorrect.
Reboxetine is rapidly absorbed after oral doses. It is mainly metabolised in the liver and has an elimination half-life of about 13 hours. In clinical trials, reboxetine has been shown to be as effective as the SSRIs in the treatment of depression but, unlike the SSRIs, reboxetine does not inhibit any of the cytochrome P450 enzymes in the liver.
The correct answer is: It does not affect hepatic cytochrome P450 enzymes.

Question 29
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 041
An orally administered medication requires five half-life periods to reach a steady state concentration. Which of the following statements is most likely to be correct with respect to this drug?

Select one:
- Doses up to five times the normal are non-toxic
- It follows first order kinetics
- A loading dose must be provided for adequate therapeutic activity
- Its rate of metabolism is independent of its concentration
- This medication must be administered only once a day

Your answer is incorrect.
Under conditions of first-order kinetics, given that a drug is rapidly and evenly distributed throughout the body and no loading dose is given, at least 5 half-lives are required to achieve a steady state concentration. Drugs following zero order kinetics show a concentration-independent elimination.
The correct answer is: It follows first order kinetics

Question 30
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 042
Which of the following is true with regard to sodium valproate?

Select one:
- Intravenous preparation is available for acute mania
- It is excreted unchanged via kidney
- It has no significant interactions with other mood stabilisers

- Should not be combined with antidepressants
- It increases carbamazepine levels

Your answer is incorrect.

Valproate appears to inhibit the glucuronidation of carbamazepine, leading to an increase in its peak levels. This may be associated with the clinically significant dose related effects of carbamazepine such as ataxia. Interactions are also seen with other psychotropics (especially lamotrigine). A study involving the administration of 100 to 300 mg/ day of chlorpromazine to schizophrenic patients already receiving valproate (200 mg BID) revealed a 15% increase in trough plasma levels of valproate.

The correct answer is: It increases carbamazepine levels

Question 31
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 043
Regarding pharmacokinetics and properties of flumazenil, which of the following is true?

Select one:
- It is an inverse agonist at GABA receptor
- It has an elimination half-life of 4 hours
- It reduces panic attacks instantaneously
- It is an epileptogenic drug
- It is a partial agonist at GABA receptor

Your answer is incorrect.

Flumazenil is a competitive antagonist of benzodiazepines at benzodiazepine sites of GABA-A receptors. Hence it is an epileptogenic drug. It is an imidazobenzodiazepine derivative with weak partial agonist action in some animal models but has no agonist activity in humans. By blocking GABA A neurotransmission, it can induce panic attacks. This has some diagnostic value too. It has a very short elimination t O and so must be given as a continuous infusion. It is 40 to 50 percent protein bound with significant tissue uptake. It is primarily metabolized by the liver with t1/2 of 40 - 80 minutes.

The correct answer is: It is an epileptogenic drug

Question 32
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 044
Which of the following is hydroxylated as one of the steps in dopamine synthesis?

Select one:
- Tryptophan hydroxylase
- L-tyrosine
- L-dopa
- L-tryptophan
- Phenylalanine hydroxylase

Your answer is incorrect.

Tyrosine is converted to dopamine by the enzymes tyrosine hydroxylase (TH) and L-amino acid decarboxylase (AADC) also called dihydroxyphenylalanine (DOPA) decarboxylase (DDC).

The correct answer is: L-tyrosine

Question 33
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 045
A 63-year-old man has a long-standing history of mood disorder. He is also taking Ramipril for hypertension. Which of the following drugs is most likely to cause a possible interaction?

Select one:
- Venlafaxine
- Amitriptyline
- Fluoxetine
- Lithium
- Bupropion

Your answer is incorrect.

Ramipril is an Angiotensin converting enzyme inhibitors (ACEI). ACEI can reduce thirst, which can lead to mild dehydration, and increase renal sodium loss, leading to increased sodium reabsorption by the kidney causing an increase in lithium plasma levels and, therefore, precipitate lithium toxicity. ACE inhibitors can also precipitate renal failure. Therefore, if ACEI are co-prescribed with lithium, more frequent monitoring of e-GFR and plasma lithium is required. (Ref: The Maudsley prescribing guidelines- 11th edition-Pg. 164)

The correct answer is: Lithium

Question 34
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 046
Which of the following is correct with regard to the pharmacology of lithium?

Select one:
- Lithium is unsafe in the elderly
- Lithium is contraindicated in patients with sick sinus syndrome
- Obesity is associated with lower rates of lithium clearance
- It is not excreted in breast milk

- Lithium cardiotoxicity is more prevalent in people on a high-salt diet

Your answer is incorrect.

Lithium is safe and effective in the elderly; it is excreted in the breast milk and must be carefully considered in lactating mothers.BMI does not affect lithium clearance as long as renal function is normal; Lithium can cause a shift in intracellular potassium and is contraindicated in sick sinus syndrome. The cardiotoxicity is more common in those on low salt diet and those with renal insufficiency.

The correct answer is: Lithium is contraindicated in patients with sick sinus syndrome

Question 35

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 041

What part of the Weschler's Adult Intelligence Scale is most likely to be resistant to brain damage?

Select one:

- Digit Symbol
- Digit Span
- Picture completion
- Object assembly
- Matrix assembly

Your answer is incorrect.

Tests on WAIS are either 'hold' or 'no hold'. The so called 'hold tests' in WAIS are thought to reflect the use of old knowledge and are relatively resistant to the effects of brain damage and ageing. These include vocabulary, information, object assembly, and picture completion. 'No hold' tests require speed of response, working memory, or the creation of new relations between unrelated items. These are more likely to show an early decline with ageing and cognitive impairment. These include digit symbol, digit span, similarities, and block design. This explains why WAIS is not the best test to measure premorbid intelligence. A deterioration index is calculated using the composite scores of HOLD and NON-HOLD subtests.

The correct answer is: Picture completion

Question 36

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 048

A 43-year-old man with pure word deafness is likely to show which of the following?

Select one:

- Reading is unimpaired
- Comprehension of speech is intact
- Writing is impaired
- Meaning of words is intact when asked
- All of the above

Your answer is incorrect.

Pure word deafness: Patient can speak read & write fluently, but comprehension is impaired only for spoken language. Bilateral (or left sided with disrupted connections to non-dominant circuit) damage to the superior temporal pole is suspected. In Wernicke's aphasia, the comprehension is impaired for both spoken and written language.

The correct answer is: Reading is unimpaired

Question 37

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 050

Which one of the following tests are not correctly linked with the main function tested?

Select one:

- Benton test- Visual memory.
- Paired associates test-verbal memory
- Stroop test-executive functioning
- Halstead-Reitan test- Intelligence
- Rey-Osterreith test- To detect prefrontal lobe pathology

Your answer is incorrect.

Rey-Osterreith test measures the recall of learned material. It is a test of visual memory. Paired associate test tests for verbal memory using paired and unpaired words. Wisconsin card sorting test is used to detect prefrontal lobe pathology.

The correct answer is: Rey-Osterreith test- To detect prefrontal lobe pathology

Question 38

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 051

There is a time gradient notable when patients develop retrograde amnesia, so that recent memories are more likely to be lost than the more remote memories. This is called

Select one:

- Interference theory
- Ribot's Law
- Law of effect
- Yerkes Dodson law
- Decay theory

Your answer is incorrect.

Ribot in 1881 wrote a book (Les Maladies de la Memoire (The Diseases of Memory)) in which he claimed on the basis of clinical observations, that brain injury affects premorbid memories in the reverse order of their formation such that newly formed memories (i.e., those formed just prior to brain injury) are impaired to a greater extent than older memories. Today, Ribot's law is referred to as temporally graded retrograde amnesia, and much evidence involving both human and animal subjects supports its validity (Ref: Wixted, J. T. (2004). On Common Ground: Jost's (1897) law of forgetting and Ribot's (1881) law of retrograde amnesia. Psychological review, 111(4), 864.)

The correct answer is: Ribot's Law

Question 39

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 052

The most obviously affected neuropsychological measure in those with post concussion syndrome is

Select one:

- Speed of processing
- Procedural memory
- Block designing
- Verbal fluency
- Picture matching

Your answer is incorrect.

Processing speed is the most obviously affected neuropsychological measure in those with post concussion syndrome.

The correct answer is: Speed of processing

Question 40

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 053

A patient has with UMN lesion of cranial nerves 9, 10, 11 and 12. Which of the following dysfunction is likely to be seen?

Select one:

- Facial droop
- Tongue fasciculations
- Swallowing problems
- Chewing problems
- Loss of touch sensation on left side of face

Your answer is incorrect.

Swallowing - IX and X; Chewing and facial sensation: - V. Facial muscle tone- VII. Fasciculations are seen in LMN lesion

The correct answer is: Swallowing problems

Question 41

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 054

You want to measure the impairment of everyday memory functioning to facilitate rehabilitation of individuals with acquired brain damage. The most suitable test is

Select one:

- Wechsler memory scale
- Halstead-Reitan Neuropsychological Battery
- The Rivermead behavioral memory test
- Ravens progressive matrices
- Thematic apperception test

Your answer is incorrect.

The Rivermead Behavioral Memory Test detects impairment of everyday memory functioning and also monitors changes following the treatment for memory difficulties. The aim of the test is to provide task analogues of everyday memory problems that appear challenging for patients with memory impairment. It can be used in adults with memory problems; individuals with traumatic or acquired brain injury, dementia, or MS.

The correct answer is: The Rivermead behavioral memory test

Question 42

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 055

A 45-year-old gentleman has history of alcohol dependence. He has no memeory problems or neurological signs but appears disorganised in day to day tasks. If a neuropsychologist tests him he is most likely to show impaired performance in which of the following tests?

Select one:

- Orientation
- Clock drawing test
- Trail making tests
- Digit span
- Picture completion test

Your answer is incorrect.

Trial making tests is a test of frontal lobe function. Patients with alcohol dependence have decrements in executive function (frontal lobe), and it could be used.

The correct answer is: Trail making tests

Question 43
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 056
A 76-year-old man was brought to Emergency Unit by his wife. He had breakfast that morning and appeared unable to remember that he and his wife have planned a day out. He repeatedly enquired where she was taking him. She reports that he appeared bewildered, but was alert and had no sensory and motor deficits throughout. He returned to normal after 10 hours. What is the most likely diagnosis?

Select one:
- Creutzfeldt-Jakob disease
- Transient global amnesia
- Alzheimer's disease
- Normal-pressure hydrocephalus
- Parkinson's disease

Your answer is incorrect.

Transient global amnesia (TGA) refers to an episode of complete and reversible anterograde and retrograde memory loss lasting up to 24 h. Patients have a persistent loss of memory for the time of the attack. During the episode, patients often appear bewildered and ask repeated questions.

The correct answer is: Transient global amnesia

Question 44
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 048
Which of the following cannot be termed as a reliable endophenotype for schizophrenia?

Select one:
- Glial cell changes
- Working memory defects
- Prepulse inhibition
- Smooth pursuit defects
- Negative symptoms

Your answer is incorrect.

The suggested criteria that define an endophenotype include: the marker is associated with the illness in the population, heritable, state-independent, co-segregates with the illness within families and is present at a higher rate in unaffected family members of the proband compared with the general population. Clinical symptom clusters do not fit these criteria.

The correct answer is: Negative symptoms

Question 45
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 050
Which one among the following is the least likely statement about Huntington's disease?

Select one:
- Huntington genes code for production of protein called huntingtin
- CAG repeat size and age are inversely correlated
- Late onset Huntington's disease is usually inherited from mother
- Paternally derived gene may be more unstable than maternally derived gene
- Normal Huntingtin gene has 35-45 CAG repeats

Your answer is incorrect.

Huntington's disease is a fully penetrant autosomal disorder. The affected gene, at 4p16, codes for huntingtin. In Huntington's disease, there are increased CAG repeats. Normal repeat lengths range from about 7 to 28 triplets. Individuals with 29-35 triplets will not develop HD but may pass an expanded allele to an offspring, while individuals with 40 or more triplets will develop HD. (Ref: New Oxford Textbook- 372)

The correct answer is: Normal Huntingtin gene has 35-45 CAG repeats

Question 46
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 051
Where does transcription take place in human cells?

Select one:
- Rough endoplasmic reticulum
- Smooth endoplasmic reticulum
- Cytoplasm
- Golgi cmplex
- Nucleus

Your answer is incorrect.

Active transcription units are clustered in the nucleus, in discrete sites called transcription factories or euchromatin.

The correct answer is: Nucleus

Question 47
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 052
Probandwise concordance rate is calculated as

Select one:
- Number of total pairs divided the affected population

- Number of twin pair divided by the total number of sib pairs
- Number of affected twins divided by total number of co-twins
- Number of twin pairs divided by the total relatives
- Number of twin pair divided by the total population

Your answer is incorrect.

A pairwise concordance rate is estimated as the number of twin pairs who both have the disorder divided by the total number of pairs. However, where there has been systematic ascertainment, one can report a probandwise concordance rate that is calculated as the number of affected twins divided by the total number of cotwins.

The correct answer is: Number of affected twins divided by total number of co-twins

Question 48

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 053

Which of the following gene, when knocked out, produces mice with phenotype resembling narcolepsy?

Select one:

- Neuropeptide Y
- Vasopressin
- Orexin
- Substance P
- Hydrocortisone

Your answer is incorrect.

Hypocretin-1 and -2 (also called orexin-A and -B) are newly discovered neuropeptides processed from a common precursor, preprohypocretin. Hypocretin-containing cells are located exclusively in the lateral hypothalamus, with widespread projections to the entire neuraxis.The discovery that hypocretins are involved in narcolepsy, a disorder associated with excessive daytime sleepiness, cataplexy and unusually rapid transitions to rapid-eye-movement sleep, opens a new field of investigation in the area of sleep control physiology (Hypocretin/orexin, sleep and narcolepsy, http://www.ncbi.nlm.nih.gov/pubmed/11340621).

The correct answer is: Orexin

Question 49

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 054

The technique used to assemble correct DNA fragments from a gene library into a continuous overlapping stretch, which will contain genes of interests is;

Select one:

- Physical mapping
- Polymerase chain reaction
- Association studies
- Mutagenesis
- Linkage analysis

Your answer is incorrect.

Physical maps of DNA give the physical, DNA-base-pair distances from one landmark to another. By using the overlapping DNA sequences at the ends of the genome pieces to keep track of where the pieces came from, physical mapping is accomplished.

The correct answer is: Physical mapping

Question 50

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 056

Polyploidy refers to which of the following phenomenon?

Select one:

- Fusion of male and female gametes producing 2n chromosome numbers in an embryo
- An inversion mutation that includes centromere of chromosomes
- Clubbing together of mitochondria and other cell organelles during interphase
- Presence of one or more extra sets of chromosomes
- Reduplicated nuclear material before cell division

Your answer is incorrect.

Polyploidy refers to the presence of extra sets of chromosomes. Triploidy refers to 3n numbers; tetraploidy refers to 4n numbers. Fusion of male and female gametes producing 2n chromosome numbers in an embryo is the normal diploid state seen in human cells.

The correct answer is: Presence of one or more extra sets of chromosomes

Question 51

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 057

A 48-year-old man has dementia. Which gene could be responsible for this condition?

Select one:

- COMT
- Progranulin
- Presenilin
- PARK-2
- Dysbindin

Your answer is incorrect.

The commonest cause of early onset dementia (<65 years) is Alzheimer's dementia. It is caused by a mutation in one of at least three genes: presenilin 1, presenilin 2 and amyloid precursor protein.

The correct answer is: Presenilin

Question 52

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 058

Recombination occurs during which phase of meiosis?

Select one:

- Prophase 1
- Anaphase 1
- Interphase
- Telophase 1
- Metaphase 1

Your answer is incorrect.

Chromosomal crossover is an exchange of genetic material between homologous chromosomes. It is one of the final phases of genetic recombination, which occurs during prophase 1 of meiosis in a process called as synapsis. Synapsis is not completed until near the end of prophase I. Crossover usually occurs when matching regions on matching chromosomes break and then reconnect to the other chromosome.

The correct answer is: Prophase 1

Question 53

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 059

Ribosome is the site of

Select one:

- Posttranslational modification of protein
- DNA storage
- Protein synthesis
- Synthesis of DNA
- Energy production via respiratory cycle

Your answer is incorrect.

Translation is carried out by more than one ribosome simultaneously.

The correct answer is: Protein synthesis

Question 54

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 060

A 53-year-old gentleman presents to the memory clinic with history of progressive short-term memory loss, which was gradual in onset. His MMSE score was 17/30. The gene, which is most likely involved in this condition, is

Select one:

- ApoE4
- Neuronal sortilin related receptor SORL1
- PSEN-1
- Dysbindin
- DISC-1

Your answer is incorrect.

The diagnosis is Alzheimer's disease (early-onset). AD is caused by mutations in the Presenilin-1 (aka PSEN) gene. Mutations in the PSEN1 gene are the most common cause of early-onset AD, accounting for up to 70% of cases

The correct answer is: PSEN-1

Question 55

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 017

Which of the following brain regions is supplied by the anterior cerebral artery?

Select one:

- cuneus
- Inferior temporal gyrus
- Lingual gyrus
- Genu of corpus callosum
- planum temporale

Your answer is incorrect.

MCA supplies blood to the cortical areas involved in speech, swallowing and language, including the lateral motor strip, lateral sensory strip, Broca's area, Wernicke's area, Heschl's gyrus, planum temporale and the angular gyrus. In addition, it provides most of the blood supply to the corpus striatum. PCA supplies the cuneus, inferior temporal gyrus and lingual gyrus. Genu and 3/4ths of corpus callosum are supplied by ACA.

The correct answer is: Genu of corpus callosum

Question 56
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 021
Which region of cerebral blood flow correlates with hallucinations and delusions?

Select one:
- Right medial temporal cortex
- Left medial temporal cortex
- Caudate nucleus
- Left prefrontal cortex
- Right prefrontal cortex

Your answer is incorrect.

Patients with schizophrenia who experience frequent auditory hallucinations show reduced grey matter density and neuronal activation. Similarly, f-MRI images have shown reduced activation in the left temporal lobe and anterior cingulate cortex. The brain structural and functional abnormalities seem to converge in the superior gyrus of the left temporal lobe.

The correct answer is: Left medial temporal cortex

Question 57
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 022
The role of ependymal cells is

Select one:
- Lining the walls of the ventricular systems
- Nutrition of neurons
- Produce myelin sheaths
- Clear neuronal debris
- Maintains the blood brain barrier

Your answer is incorrect.

Ependymal cells are a special type of glial cells that covers the lining in the walls of the ventricular systems and enable CSF circulation.

The correct answer is: Lining the walls of the ventricular systems

Question 58
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 023
The disengagement of attention is mediated by

Select one:
- Medial temporal cortex
- Medial parietal cortex
- Lateral temporal cortex
- Prefrontal cortex
- Occipital cortex

Your answer is incorrect.

The parietal lobe is postulated to be involved in engagement and disengagement of attention. In conditions like Alzheimer's disease, the medial parietal cortex is affected in early stages, and attentional deficits are seen before language and visuospatial deficits.

The correct answer is: Medial parietal cortex

Question 59
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 024
Which neuroglial cells act as scavenger cells at the sites of CNS injury?

Select one:
- Schwann cells
- Microglia
- Satellite cells
- Astrocytes
- Ependymal cells

Your answer is incorrect.

Microglia acts as scavenger cells at sites of CNS injury; Ependymal cells aids the flow of CSF. Schwann cells are responsible for the myelin sheath formation in peripheral nervous system (neurilemma formation). Satellite cells provide support to neurons in sensory and autonomic ganglia.

The correct answer is: Microglia

Question 60
Not answered
Marked out of 1.00
Flag question

MoK Neurochemistry 021
Which of the following enzymes is seen only in the neuron and not in the glial cells?

Select one:
- Glutamic Acid Decarboxylase

- monoamine oxidase
- dopadecarboxylase
- cholinesterase
- Glutamine Synthase

Your answer is incorrect.

Glutamic acid decarboxylase [GAD] is a neuron-specific enzyme that is not found in glial cells. Hence, antibodies to GAD can be used to identify neurons that release GABA. Glutamine synthase is a glial enzyme that metabolizes glutamate into glutamine.

The correct answer is: Glutamic Acid Decarboxylase

Question 61

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 022
MAO-A is available primarily in which of the following body tissues?

Select one:

- Spleen
- Lungs
- Brain
- Kidney
- Gut

Your answer is incorrect.

Monoamine oxidases are of two types: the brain predominantly has monoamine oxidase inhibitor -B and in the liver/GI tract, predominantly monoamine oxidase inhibitor-A is noted.

The correct answer is: Gut

Question 62

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 023
The most common side effect when using sildenafil is:

Select one:

- Headache
- Insomnia
- Double vision
- Hypotension
- Bradycardia

Your answer is incorrect.

Viagra is an indirect vasodilator, resulting in headaches

The correct answer is: Headache

Question 63

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 024
Low CSF concentration of 5-Hydroxy Indole Acetic Acid is associated with which of the following?

Select one:

- Cheese Reaction
- Serotonin Syndrome
- Impulsivity
- Visual Hallucinations
- Coronary Artery Disease

Your answer is incorrect.

Animal and human research suggests that the central serotonin system is involved in the inhibition of impulsive behaviour A low CSF 5-HIAA concentration was primarily associated with impulsivity. http://archpsyc.ama-assn.org/cgi/content/abstract/51/1/20

The correct answer is: Impulsivity

Question 64

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 025
Nitric oxide is synthesized from which of the following?

Select one:

- Toluene
- Benzene
- Phosphokinase
- L-arginine
- L-tryptophan

Your answer is incorrect.

Nitric oxide is synthesized from L-arginine when stimulated by nitric oxide synthase.

The correct answer is: L-arginine

Question 65

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 013

Which of the following human spongiform encephalopathy has the slowest course?

Select one:

- Sporadic CJD
- Variant CJD
- Iatrogenic CJD
- Gerstmann- Straussler- scheinker syndrome
- Kuru

Your answer is incorrect.

Kuru is a type of prion disease which has the slowest course. The incubation period was prolonged- up to 40 years before disease onset, which was then rapid and fatal. It is thought to have been transmitted by ritual cannibalism. Symptoms include personality and behavioural disturbances, progressive cerebellar ataxia, dementia and death.

The correct answer is: Kuru

Question 66

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 014

Mr. X is a 73-year-old gentleman with 5-year history of Parkinson's disease, cognitive decline and visual hallucinations. Post mortem evidence suggests alpha syn-nuclein deposits and cytoplasmic inclusions. The common microscopic finding expected in this case would be

Select one:

- Pick cells
- Tau proteins
- Hirano bodies
- Lewy bodies
- Pick bodies

Your answer is incorrect.

The clinical features, in this case, are suggestive of Parkinson's disease dementia. In Parkinson's disease, the Lewy bodies are confined to the substantia nigra. The Lewy bodies are weak eosinophilic, spherical, cytoplasmic inclusions. Parkinson's disease and Lewy body dementia contains an accumulation of alpha-synuclein and is classified under synucleinopathies.

The correct answer is: Lewy bodies

Question 67

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 018

Which of the following neuroimaging finding is most suggestive of Alzheimer's disease?

Select one:

- Occipital hypoperfusion on HMPAO- SPECT
- Impaired dorsolateral cortex function
- Activation of primary auditory cortex on functional MRI
- Increased signal in the Pulvinar nucleus of thalamus bilaterally
- Medial temporal atrophy on structural MRI scan

Your answer is incorrect.

MRI often shows atrophy of grey matter involving the hippocampus, entorhinal cortex and amygdala in the medial temporal lobe.

The correct answer is: Medial temporal atrophy on structural MRI scan

Question 68

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 013

Which of the following neural structures is strongly implicated in the regulation of P50 suppression, a proposed endophenotype in schizophrenia?

Select one:

- Frontal cortex
- Angular gyrus
- Basal ganglia
- Hippocampus
- Ventral tegmentum

Your answer is incorrect.

P50 suppression is regulated by wide-ranging neural circuitry, prominently involving hippocampal structures.

The correct answer is: Hippocampus

Question 69

MoK Neurophysiology 014

Not answered

Marked out of 1.00

Flag question

A 56-year-old lady was admitted with depression following a suicide attempt by self-induced skin laceration. She was increasingly isolating herself and complained of somatic pain symptoms in her back. She also had constipation for few weeks. No past psychiatric history was noted. Routine blood tests revealed an isolated anomaly of high serum calcium level (15.8 mg/dL). Which of the following is the most likely diagnosis?

Select one:

- Chronic renal failure
- Hyperparathyroidism
- Hypothyroidism
- Hyperprolactinemia
- Hypoparathyroidism

Your answer is incorrect.

Primary hyperparathyroidism is generally related to parathyroid adenomas. Most patients are asymptomatic but can have recurrent kidney stones, peptic ulcers, dehydration, constipation, and altered mental status. Diagnosis can be confirmed with measurement of parathyroid hormone levels. (Ref: http://psy.psychiatryonline.org/cgi/content/full/40/6/486)

The correct answer is: Hyperparathyroidism

Question 70

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 015

Which of the following statements with regard to alpha waves in the EEG is correct?

Select one:

- Antipsychotics abolish alpha waves
- It is maximal over occipital region
- It reduces with eye closure
- Excess of alpha rhythm is seen in Huntington's disease
- It is a higher frequency band than beta waves

Your answer is incorrect.

The alpha brain waves oscillate about 10 times per second, and the range is 8-13 cycles per second. The brain waves called 'Alpha' were the first to be discovered (around 1908, by an Austrian Psychiatrist named Hans Berger).Neuroleptics commonly produce increased alpha rhythm without obvious clinical manifestations. In Huntington's the EEG will be low voltage, with poorly formed or absent alpha waves.

The correct answer is: It is maximal over occipital region

Question 71

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI023

Blood supply to the brain

Match the specific areas of the brain with their blood supply from the list provided:

Most of cerebellar hemisphere — Choose...

Posterior limb of internal capsule — Choose...

Cerebellum-vermis — Choose...

Your answer is incorrect.

Most of the cerebellar hemispheres including superior and middle cerebellar peduncles and cerebellar nuclei are supplied by the superior cerebellar artery. The vermis of the cerebellum is supplied by the posterior inferior cerebellar artery.

The posterior limb of the internal capsule is supplied by the posterior cerebral artery.

The correct answer is: Most of cerebellar hemisphere – Superior cerebellar artery, Posterior limb of internal capsule – Posterior cerebral artery, Cerebellum-vermis – Posterior inferior cerebellar artery

Question 72

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI024

Neurological symptoms

Find one term each to describe the following presentations:

A 65-year-old gentleman was unable to identify familiar objects such as key or coins by touch with eyes closed. He has difficulty in describing their shape and texture. However, he identifies them when eyes are open. — Choose...

A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing — Choose...

A man has left side hemiplegia after a stroke. He is unaware of his deficit. — Choose...

Your answer is incorrect.

Agraphaesthesia is defined as inability to recognize what number or alphabet is scratched on one's skin without seeing. **Astereognosis** is defined as inability to recognize objects by palpation, and without visual inspection. **Anosognosia** refers to the inability to identify one's neurological deficits.

The correct answer is: A 65-year-old gentleman was unable to identify familiar objects such as key or coins by touch with eyes closed. He has difficulty in describing their shape and texture. However, he identifies them when eyes are open. – Astereognosis, A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing – Agraphaesthesia, A man has left side hemiplegia after a stroke. He is unaware of his deficit. – Anosognosia

Question 73
Not answered
Marked out of 5.00
Flag question

MoK Neurosciences EMI026
Receptor localization
For each of these areas of the brain, identify the main neurotransmitter pathways localized to the specified regions:

Median raphe nuclei: Choose...
Nucleus basalis of Meynert: Choose...
Substantia nigra: Choose...
Dorsal raphe nuclei: Choose...
Locus coeruleus: Choose...

Your answer is incorrect.

The major site of serotonergic cell bodies is in the upper pons and midbrain - in the median and dorsal raphe nuclei.

A group of acetylcholine producing neurons in the nucleus basalis of Meynert project to the cerebral cortex, limbic system, hypothalamus and thalamus.

The major concentration of norepinephrine is in the cell bodies that project upward in the brain is in the locus coeruleus in the pons.

Dopamine mainly localized in nigrostriatal pathways, mesolimbic and mesocortical pathways and tuberoinfundibular pathways.

The correct answer is: Median raphe nuclei – Serotonin, Nucleus basalis of Meynert – Acetylcholine, Substantia nigra – Dopamine, Dorsal raphe nuclei – Serotonin, Locus coeruleus – Norepinephrine

Question 74
Not answered
Marked out of 5.00
Flag question

MoK Neurosciences EMI028
Lobar dysfunction
For each of the clinical descriptions given below, identify the area of the brain damaged.

Alexia without agraphia, colour agnosia and visual object agnosia: Choose...
Profound changes in personality, problems with verbal expression and forced utilization behaviour: Choose...
Profound amnesia: Choose...
Visuospatial agnosia, prospagnosia and complex visual hallucinations: Choose...
Dressing apraxia, prospagnosia, hemisomatognosia: Choose...

Your answer is incorrect.

Lesions in the dominant occipital lobe result in alexia without agraphia, colour agnosia and visual object agnosia.

Lesions in the non-dominant occipital lobe result in visuospatial agnosia, metamorphopsia (image distortion) and complex visual hallucinations.

Damage to the frontal lobes results in a change in one's personality, with poor motivation, poor judgement and impaired social awareness. If the orbital frontal region (just above the eye sockets) is damaged, forced utilization behaviour may be seen. Here, when objects are placed in front of the subject, they will be picked up and used appropriately even when the subject is told not to do so.

Profound amnesia results from bilateral medial temporal lobe lesions. The causes of such lesions would include infections, tumour and epilepsy. Both anterograde and retrograde memory will be impaired with intact immediate memory (i.e. unimpaired digit span) and preserved executive task performance.

Lesions in the non-dominant parietal lobe result in anosognosia, hemisomatagnosia, dressing apraxia and prosopagnosia. Gerstmann syndrome is associated with lesions of the dominant parietal lobe.

(Ref: Core Psychiatry: pg 16-20)

The correct answer is: Alexia without agraphia, colour agnosia and visual object agnosia – Dominant occipital lobe, Profound changes in personality, problems with verbal expression and forced utilization behaviour – Frontal lobe, Profound amnesia – Bilateral temporal lobe lesions, Visuospatial agnosia, prospagnosia and complex visual hallucinations – Non-dominant occipital lobe, Dressing apraxia, prospagnosia, hemisomatognosia – Non-dominant parietal lobe

Question 75
Not answered
Marked out of 3.00
Flag question

MoK Pharmacology EMI013
Pharmacodynamics
Select one mechanism of action each for the following drugs

Rivastigmine: Choose...
Selegeline: Choose...
Diazepam: Choose...

Your answer is incorrect.

Rivastigmine is a cholinesterase Inhibitor. It acts by inhibiting acetylcholinesterase enzyme that breaks down acetylcholine centrally. Rivastigmine inhibits both the acetyl and butyl-cholinesterase while donepezil and galantamine are acetyl specific.

Selegiline is a monoamine oxidase Inhibitor - selective for the enzyme MAO-B at normal therapeutic doses; selectivity is lost when the drug is administered (usually as a patch) at higher doses, leading to some antidepressant action.

Diazepam is a type of benzodiazepine which acts via a specific site called omega site in GABA-A complex. All benzodiazepines are full agonists except clonazepam which is a partial agonist. They facilitate GABA action on GABA-A complex - thus inhibitory neurotransmission via chloride ion flow is facilitated. Benzodiazepines have no direct agonistic action in the absence of GABA.

The correct answer is: Rivastigmine – Butyryl-cholinesterase inhibitor, Selegeline – MAO-B Inhibitor, Diazepam – GABA-A agonist

Question 76

MoK Pharmacology EMI016

Not answered

Marked out of 3.00

Flag question

Pharmacokinetics (1)
Match each characteristic below with the most likely drug from the list given

This drug undergoes no metabolism, and it is excreted unchanged by the kidney — Choose...

This drug accelerates its own metabolism and the metabolism of other drugs by inducing the hepatic microsomal enzymes — Choose...

The metabolism of this drug follows zero order kinetics — Choose...

Your answer is incorrect.

Lithium is orally well absorbed but not metabolised by the liver; it is renally excreted.

Carbamazepine is metabolized by the hepatic cytochrome P450 2D6 system, which it also robustly induces. As a result of this autoinduction, the rate of metabolism of carbamazepine (and other P450 substrates) usually increases over the first several weeks of treatment. The initial steady state may be attained within 4 to 5 days, but autoinduction may delay final steady state until 3 to 4 weeks after treatment initiation. Hence, the level of carbamazepine must be monitored, and its dose often must be raised during this early phase of treatment.

Slow release preparations e.g. lithium MR, depot preparations follow zero-order absorption kinetics; drugs that rapidly saturate enzymes such as alcohol and phenytoin follow zero-order elimination kinetics

The correct answer is: This drug undergoes no metabolism, and it is excreted unchanged by the kidney – Lithium, This drug accelerates its own metabolism and the metabolism of other drugs by inducing the hepatic microsomal enzymes – Carbamazepine, The metabolism of this drug follows zero order kinetics – Phenytoin

Question 77

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology CXEMI017

Drug interactions (1)
Choose agents that will have clinically significant interaction for each description below

1. A 45-year-old patient is an enthusiastic naturalist and is keen on drinking grapefruit juice for breakfast. (Choose TWO)

2. A 35-year-old patient is on long-term NSAID therapy for an ankle injury. (Choose ONE)

Explanation

The "grapefruit juice effect" is due to components of grapefruit juice (bergamottin, 6, 7-dihydroxybergamottin, and naringenin - these significantly increase drug oral bioavailability by selectively and rapidly down-regulating intestinal (but not liver) CYP3A4 and to a lesser extent, 1A2. This effect is greatest in drugs with high first-pass metabolism such as the dihydropyridine calcium antagonists, felodipine and nimodipine, terfenadine, carbamazepine, triazolam and midazolam (to some extent diazepam), simvastatin and methylprednisone. Buspirone and Pimozide are also significantly affected by grapefruit. (Ref: Psychosomatics; 2005: 46). The class of medications that could increase serum lithium levels and precipitate lithium toxicity includes diuretics (Loop and Thiazide diuretics), NSAIDs and ACE inhibitors.

The correct answer is:

A 45-year-old patient is an enthusiastic naturalist and is keen on drinking grapefruit juice for breakfast. (Choose TWO) - Carbamazepine, Pimozide,

A 35-year-old patient is on long-term NSAID therapy for an ankle injury. (Choose ONE) - Lithium

Question 78

Not answered

Marked out of 5.00

Flag question

MoK Pharmacology CXEMI023

Antipsychotics & side effects
For each of the following situations, identify the most likely medication associated with it

1. Constipation and paralytic ileus

2. Orthostatic hypotension (choose two answers)

3. Nausea and vomiting

4. Post-injection delirium

Explanation:

Clozapine has pronounced anticholinergic properties. It can cause severe constipation. Effective treatment or prevention of constipation is essential, as this can prove fatal by progressing to paralytic ileus and bowel ischemia, followed by sepsis and perforation.

Clozapine and Risperidone can cause severe orthostatic hypotension due to alpha one adrenergic blockade.

Nausea, vomiting and gastric side effects are common with Aripiprazole due to 5-HT1A receptor partial agonistic actions.

Olanzapine if administered as long-acting depot injection (Olanzapine Pamoate) could produce symptoms suggestive of antipsychotic overdose in a small number of subjects (< 0.1%), a phenomenon that has been termed "post-injection delirium/sedation syndrome" (PDSS). PDSS is characterized by symptoms related to excessive sedation and delirium soon after the injection, consistent with some of the symptoms of oral olanzapine overdose.

Ref: McDonnell et al., BMC Psychiatry. 2010; 10: 45. Retrieved from http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2895590/

The correct answer is:
Constipation & Paralytic Ileus - Clozapine,

Orthostatic Hypotension - Clozapine,Risperidone

Post-injection delirium - Olanzapine,

Nause and Vomiting - Aripiprazole

Finish review